

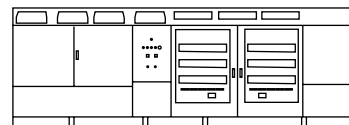


PCSM

IEC

Easy maintenance.
Integrated MV solution in the same enclosure.
Advanced grid support.
Compatible with all battery technologies.





REFERENCES		FP4200MH	FP4207MH	FS4208MH
AC	AC Output Power (kVA/kW) @40 °C ^[1]	4200		
	AC Output Power (kVA/kW) @50 °C ^[1]	3900		
	Operating Grid Voltage (kV)	34.5 kV ±10%	33 kV ±10%	
	Operating Grid Frequency (Hz)	60 Hz	50 Hz	
DC	DC Voltage Range ^[3]	934 V - 1500 V		
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η) (preliminary)	97.80% including MV transformer		
	Euroeta (η) (preliminary)	97.51% including MV transformer		
ENVIRONMENT	Max. Altitude (above sea level) ^[5]	2000 m		1000 m
REFERENCES		FP4105MH	FP4010MH	
AC	AC Output Power (kVA/kW) @40 °C ^[1]	4105	4010	
	AC Output Power (kVA/kW) @50 °C ^[1]	3810	3720	
	Operating Grid Voltage (kV)	34.5 kV ±10%		
	Operating Grid Frequency (Hz)	60 Hz		
DC	DC Voltage Range ^[3]	913 V - 1500 V	891 V - 1500 V	
EFFICIENCY & AUX. SUPPLY	Efficiency (Max) (η) (preliminary)	97.76% including MV transformer	97.75% including MV transformer	
	Euroeta (η) (preliminary)	97.50% including MV transformer	97.48% including MV transformer	
ENVIRONMENT	Max. Altitude (above sea level) ^[5]	2000 m		
COMMON FEATURES				
AC	Current Harmonic Distortion (THDi)	<3% per IEEE 519		
	Power Factor (cos phi) ^[2]	0.5 leading ... 0.5 lagging		
	Reactive Power Compensation	Four quadrant operation		
	Overload Capability ^[7]	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s		
DC	Maximum DC Voltage	1500 V		
	DC Voltage Ripple	<3%		
	Max. DC Continuous Current (A)	4590		
	Max. DC Short Circuit Current (kA)	500 kA with a time constant of 1 ms		
	Battery Technology	All type of batteries (BMS required)		
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5		
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3		
	Weight (lbs)	30865		
	Weight (kg)	14000		
	Type of Ventilation	Forced air cooling		
ENVIRONMENT	Degree of Protection	IP55		
	Operating Temperature Range ^[4]	From -40 °C to +60 °C, >40 °C power derating		
	Operating Relative Humidity Range	From 4% to 100% non-condensing		
	Storage Temperature Range	From -40 °C to +60 °C		
CONTROL INTERFACE	Communication Protocol	Modbus TCP		
	Power Plant Controller	Optional. Third party SCADA systems supported.		
	Keyed ON/OFF Switch	Standard		
PROTECTIONS	Ground Fault Protection	Insulation monitoring device		
	Humidity Control	Active heating		
	General AC Protection & Disconn.	MV switchgear (2L+V)		
	General DC Protection & Disconn.	High-speed fuses, Motorized DC disconnect switches ^[6]		
	Overvoltage Protection	Type II for AC and Type I+II for DC		
STANDARDS	Safety	IEC 62109-1 / IEC 62109-2 / IEC 62477-2		

NOTES

- [1] Values at 1.00-Vac nom and cosφ=1. Consult Power Electronics for derating curves.
[2] Consult P-Q charts available: $Q(kVA) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
[3] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
[4] Optional available for temperatures below -25 °C.
[5] Consult Power Electronics for altitudes above 1000 m.
[6] Battery short circuit disconnection must be done on the battery side.
[7] Available from January 2027.